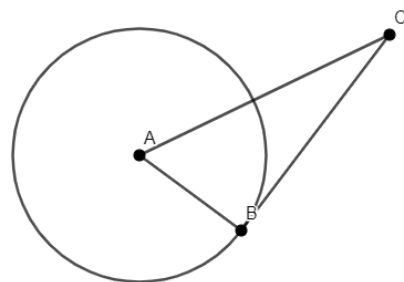


Geometry
Unit 12
Lesson 5 Practice

Name **Answer Key** _____
Date _____
Period _____

1. Circle A is shown with point C outside the circle, where point B is on the circle, $AC = 26$, $CB = 24$, and $BA = 10$. Verify if $\overline{BC}$ is tangent to circle A using Pythagorean Theorem.

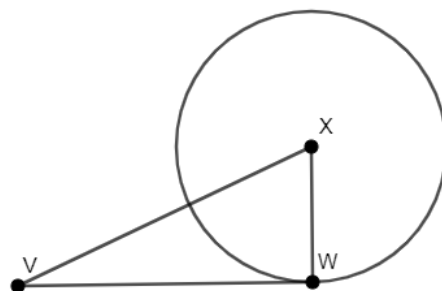
$$\begin{aligned}(24)^2 + (10)^2 &= (26)^2 \\ 576 + 100 &= 676 \\ 676 &= 676 \checkmark \Rightarrow \text{TANGENT}\end{aligned}$$



Circle X on the right has $\overline{VW}$ as a tangent segment where W is the point of tangency. $VX = 17$ and $VW = 15$.

2. What is the length of $\overline{XW}$?

$$\begin{aligned}(15)^2 + (XW)^2 &= (17)^2 \\ 225 + (XW)^2 &= 289 \\ (XW)^2 &= 64 \Rightarrow XW = 8\end{aligned}$$



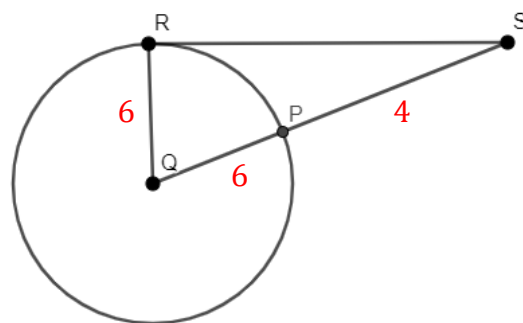
3. What is the area of $\triangle XWV$?

$$\frac{1}{2}(15)(8) = 60 \text{ units}^2$$

In Circle Q , $\overline{RS}$ is tangent to the circle at point R , $QR = 6$, and $PS = 4$.

4. What is the length of $\overline{RS}$?

$$\begin{aligned}6^2 + (RS)^2 &= (10)^2 \\ 36 + (RS)^2 &= 100 \\ (RS)^2 &= 64 \Rightarrow RS = 8\end{aligned}$$



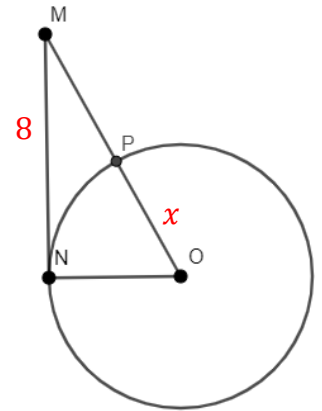
5. What is the area of $\triangle QRS$?

$$\frac{1}{2}(6)(8) = 24 \text{ units}^2$$

In Circle O , $\overline{MN}$ is a tangent segment whose point of tangency is point N . $MN = 8$ and $MP = 2$.

6. What is the length of $\overline{ON}$?

$$\begin{aligned}(2 + x)^2 - 8^2 &= x^2 \\ 4 + 4x + x^2 - 64 &= x^2 \\ 4x &= 60 \\ x &= 15 \\ ON &= 15\end{aligned}$$



7. What is the length of $\overline{MO}$?

$$\begin{aligned}2 + 15 &= 17 \\ MO &= 17\end{aligned}$$

8. What is the perimeter of $\triangle MNO$?

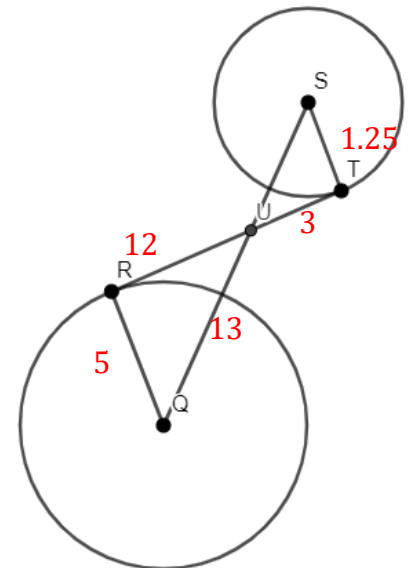
$$17 + 15 + 8 = 40 \text{ units}$$

In the figure on the right, $\overline{RT}$ is a common tangent segment of both Circle Q and Circle S . $QR = 5$, $RU = 12$, and $TU = 3$.

9. What is the length of $\overline{ST}$?

$$\frac{5}{12} = \frac{ST}{3}$$

$$15 = 12(ST) \Rightarrow \boxed{ST = 1.25}$$



10. What is the length of $\overline{QS}$?

$$\begin{aligned}(12)^2 + (5)^2 &= (QU)^2 \\ 13 &= QU\end{aligned}$$

$$\begin{aligned}(SU)^2 &= 3^2 + (1.25)^2 & \Rightarrow QS &= 13 + 3.25 \\ SU &= 3.25 & QS &= 16.25\end{aligned}$$